

Samyak Hadgekar

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PROJECTS

Patient Pain Assessment & Visualization Web Platform

Built a healthcare web system to manage patient-wise pain assessment data from a medical device, enabling real-time viewing, clinician-approved data commits, and advanced visualization through charts and pain maps using React and a PHP-MySQL backend.

Unified Multimodal AI Assistant with RAG

Built a complete local AI system combining general chat, image understanding, and document Q&A in a single interface using Phi-3 and LLaVA models via Ollama, with a Retrieval-Augmented Generation pipeline powered by LangChain, ChromaDB, and HuggingFace embeddings for PDF question-answering, and a multi-tab Gradio interface optimized to run fully offline.

Customer segmentation(Clustering) - (Streamlit App)

Developed a customer segmentation model using K-Means clustering to group users based on purchasing behavior and engagement patterns. Built and deployed an interactive Streamlit web app enabling real-time segment prediction and business insights. [Link](#)

Sales prediction model(Regression) - (Streamlit App)

Developed an end-to-end sales prediction system using XGBoost regression on the BigMart dataset to forecast product outlet sales. Implemented data preprocessing, feature encoding, and model evaluation using R^2 score, and deployed the model as an interactive Streamlit web application for real-time sales prediction. [Link](#)

Rule-Based chatbot model(NLP)

Built an NLP-based chatbot using Python, NLTK, and TensorFlow, implementing tokenization, stemming, and bag-of-words. Designed a neural network model for intent classification and real-time response generation. Trained and deployed model with Keras, achieving accurate predictions and scalable conversation handling. Built an NLP-based chatbot using Python, NLTK, and TensorFlow, implementing tokenization, stemming, and bag-of-words. Designed a neural network model for intent classification and real-time response generation. Trained and deployed model with Keras, achieving accurate predictions and scalable conversation handling.

EDUCATION

MIT-CSN

B.tech (AI & Data Science)

MAHARASHTRA, IN

2023-2027

SKILLS

- **Programming language** - C, C++, python, java.
- **Machine Learning** - Regression, Clustering, Feature Engineering, Model Evaluation (R^2 , RMSE), RAG, Multimodal AI
- **Libraries & Frameworks** - Pandas, NumPy,, Matplotlib, LangChain, Gradio, HuggingFace, Ollama, ChromaDB.
- **Deployment & Tools** - Google Colab, VS Code, GitHub, jupyter, streamlit, Ollama.
- **AI & Analytics** - Data Cleaning, Analysis & Visualization, Local LLM Deployment, Document Intelligence.

ACHIEVEMENTS & INITIATIVES

- **Selected among the Top 11 Teams Selected Nationwide** for **MEDIC 2025 – Medical Device Innovation Course** at **IIT Bombay (BETIC)**, gaining hands-on exposure to clinical problem identification, med-tech innovation, and interdisciplinary collaboration.
- **Selected in Smart India Hackathon 2024** as part of a team developing an e-learning platform aimed at **reducing college student dropout rates, shortlisted from a large national pool of entries.**
- **Designed and developed an industry-grade onion distribution device prototype** during first year of college, collaborating in a two-member team, with a focus on efficient material handling and practical industrial applicability.
- **Semi-Finalist** in a **City-Level Speech Competition** held at **SB College of Engineering**, recognized for effective public speaking and articulation of ideas.